Cars: Part I

Physics and Mathematics of Sustainable Energy

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A few facts:

- Calorific value of gasoline: 10 kWh per liter or 38 kWh per gallon .
- Average gas mileage for car in US: 25 mpg, but this ranges considerably.
- mpg = miles per gallon
- Carbon intensity of gasoline: 240g per kWh or 9kg per gallon. Note: This varies, because the composition of gasoline varies from country to country.
- Carbon intensity of electricity: https://app.electricitymaps.com/map/fifteen_minutes

1. Suppose you drive 20 miles each way to work every workday in an average US gas car.

   (a) How much gas does this use?

   (b) How much energy does this use? Answer in kWh per person per day. Is this a lot or a little?

   (c) How much carbon dioxide is emitted by the car in one year? Answer in tons per year. Is this a lot or a little?

2. Suppose you buy a 10kW (nameplate) solar PV system.

   (a) How much electricity will you generate in one year? Assume a capacity factor of 0.12.

   (b) Assuming that the solar PV saves 300 grams of emissions for every kWh, how much CO_2 will have been prevented due to your purchase of the solar PV panels?

3. Suppose you have two lights on your desk that you leave on for an average of 3 hours a day. You switch from compact fluorescent bulbs that draw 14 watts to LEDs that draw 7 watts.

   (a) About how much energy will you save in one year?

   (b) How much less CO_2 will be emitted as a result?

   (c) How far would you have to drive to emit an amount of carbon dioxide equivalent to that which you saved by switching bulbs?

4. Let's compare driving 1000 miles in conventional and electric vehicles.

   (a) In the conventional car, how much gas does this use?

   (b) How much does this gas cost?

   (c) How much CO_2 is emitted by the car?

   (d) How much of the thermal energy released when burning the gasoline goes into the kinetic energy of the car? Assume that the car's engine has an efficiency of 0.25.

   (e) How many kWh of electricity would be needed by an electric car to go 1000 miles. Assume that the efficiency of the electric car is 0.85.

   (f) How much would this electricity cost?

   (g) How much CO_2 would be emitted as a result of generating this amount of electricity, assuming a carbon intensity of 300g/kWh.

   (h) How much CO_2 would be emitted as a result of generating this amount of electricity if the electricity was generated in a coal-burning power plant with an intensity of 0.8 kg/kWh?